Date : ☐☐/☐☐/20☐☐ SEAT NO. ☐☐☐☐☐☐

201P	III	XX

(2829aa)

(E)

1 - 60

CHEMISTRY

Std. : 12^{th}

Time : 3 Hours (3 Pages) Max. Marks : 70

General Instructions :

1) All the questions are compulsory.
2) Section 'A' contains Q. No. 1 to 4 of multiple choice type of questions carrying one mark each.
   Q. No. 5 to 8 are very short answer type of questions carrying one mark each.
3) Section 'B' contains Q. No. 9 to 15 of short answer type of questions carrying two marks each.
   Internal choice is provided to only one question.
4) Section 'C' contains Q. No. 16 to 26 of short answer type questions carrying three marks each.
   Internal choice is provided to only one question.
5) Section 'D' contains Q. No. 27 to 29 of long answer type questions carrying five marks each.
   Internal choice is provided to each question.
6) Use log table is allowed. Use of calculator is not allowed.
7) Given data - Gas constant $R = 8.314\ J K^{-1} mol^{-1}$; Atomic Weight : N = 14 ; Avogadro's no. $(N_A) = 6.022 \times 10^{23}$

Section - A (1 Mark each) (8)

Q.1 Which reagent cannot be used to prepare alkyl halide from an alcohol ? 1

A) $HCl + ZnCl_2$ B) NaCl C) PCl_5 D) $SOCl_2$

Q.2 A matchbox exhibits 1

A) Cubic geometry B) Monoclinic geometry

C) Orthorhombic geometry D) Tetragonal geometry

Q.3 A reaction showing slag formation is 1

A) $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$ B) $ZnCO_3 \rightarrow ZnO + CO_2$

C) $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$ D) $FeO + SiO_2 \rightarrow FeSiO_3$

Q.4 Which is not an optically active amino acid ? 1

A) Alanine B) Glycine C) Arginine D) Lysine

Q.5 At what temperature will the following reaction have $\Delta G = -777.8$ kJ, if $\Delta H = -843.7$ kJ and $\Delta S = -165$ kJ ? 1

$2Pb_{(s)} + 3O_{2(g)} \rightarrow 2PbO_{(s)} + 2O_{2(g)}$

Q.6 For a general chemical reaction, $mA + nB \rightarrow pC + qD$. The rate law is $\frac{dx}{dt} = K[A]^x [B]^y$. 1

What is the molecularity and overall order of reaction ?

Q.7 Identify 'C' in the following reaction. 1

$$\text{Ethanol} \xrightarrow{PBr_3} A \xrightarrow{\text{alc. KOH}} B \xrightarrow[\text{ii) } H_2O,\ \Delta]{\text{i) } H_2SO_4 / \text{R.T}} C$$

Q.8 Arrange the following amines in increasing order of basicity in aqueous solution. 1

i) CH_3NH_2 ii) $(CH_3)_2NH$ iii) $(CH_3)_3N$

Section - B (2 Mark each) (14)

Q.9 Draw a neat and labelled diagram of Bessemer converter used in extraction of copper. 2

Q.10 Classify the following ligands into monodentate, polydentate and ambidentate. 2

i) NH_3 ii) ethylene diamine iii) NO_2^- iv) oxalato

OR

Q.10 Draw the geometrical isomers of $[CoCl_2(en)_2]^+$. 2

Q.11 The solubility of dissolved oxygen to 27°C is 2.6×10^{-3} mol dm^{-3} at 2 atm. Find its solubility at 8.4 atm. and 27°C. 2

Q.12 How is nylon - 66 prepared ? 2

Q.13 Draw the structure of $H_4P_2O_6$ and $H_4P_2O_7$. Find out oxidation state of phosphorus in each oxyacid. 2

Q.14 Write ozonolysis reaction for propylene. 2

Q.15 Write the cell reaction and calculate the standard potential of the cell. 2

$Ni_{(s)} | Ni^{2+}_{(1M)} \,||\, Cl^-_{(1M)} | Cl_{2\,(g,\,1\,atm)} | Pt.$ $E^\circ_{Cl_2} = 1.36$ V and $E^\circ_{Ni} = -0.25$ V.

Section - C (3 Mark each) (33)

Q.16 State Kohlrausch's law. Distinguish between electronic and electrolytic conductors. 3

Q.17 Define - Lipids. How are lipids classified ? 3

Q.18 What is freezing point of a liquid ? How is molar mass of a non-volatile solute related to the depression in freezing point ? Derive an equation for it. 3

Q.19 State the hybridization of the nitrogen atom in amines. Write a short note on Mendius reduction. 3

Q.20 What is the action of acidified potassium dichromate on - 3

a) SO_2 b) KI c) ethanol ?

Q.21 The rate constant of a first order reaction is $6.8 \times 10^{-4}\ s^{-1}$. If the initial concentration of the reactant is 0.04 M, what is its molarity after 20 minutes ? How long will it take for 25% of the reactant to react ? 3

Q.22 What are broad spectrum antibiotics ? Explain any two chemical methods of food preservation. 3

Q.23 Predict the expected product of substitution reactions. 3

a) Isobutyl chloride + Sodium ethoxide

b) n-butyl chloride + sodium

c) 1-chloropropane + aq. Potassium hydroxide

Q.24 Explain how does nitrogen exhibit anomalous behaviour amongst gr. 15 elements. 3

Q.25 How is diethyl ether prepared by continuous etherification process ? Why is the process so called ? 3

OR

Q.25 How is carbolic acid prepared from cumene ? What is the action of bromine water on carbolic acid ? 3

Q.26 2.8×10^{-2} kg of nitrogen is expanded isothermally and reversibly at 300 K from $15.15 \times 10^{5}\ Nm^{-2}$ when the work done is found to be -17.33 kJ. Find the final pressure. 3

Section - D (5 marks each) (15)

Q.27 Why does copper shows abnormal electronic configuration ? 2

An element crystallises in fcc crystal lattice and has a density of 10 g cm^{-3} with unit cell edge length of 100 pm. Calculate number of atoms present in 1 g of crystal. 3

OR

Q.27 What are chemical twins ? Give examples. 2

A metal crystallises into two cubic faces namely face centered (fcc) and body centered (bcc) whose unit cell edge lengths are 3.5A° and 3A° respectively. Find the ratio of the densities of fcc and bcc. 3

Q.28 How is orlon prepared ? 1

Write the formula of the complex Diamminedichloridoplatinum (II). 1

Explain the structure of SO_2 molecule. Write uses of SO_2. 3

OR

Q.28 How is teflon prepared ? 1

Write IUPAC name for the complex : $Na_2[Zn(CN)_4]$ 1

What is the action of chlorine on - a) hydrogen b) turpentine oil c) CS_2 ? 3

Q. 29 State and explain Hess's law of constant heat summation. 2

Write a note on : Aldol condensation. 3

OR

Q. 29 Which of the following pairs have larger entropy ? Why ? 2

a) $CO_{2(s)}$ or $CO_{2(g)}$ b) $CH_3OH_{(l)}$ or $C_2H_5-O-C_2H_{5(l)}$

How will you bring about the following conversions ? 3

a) Ethylidene dichloride to acetaldehyde.

b) Calcium acetate to acetone.

c) Cyclopropane carboxylic acid to cyclopropyl methanol.